# Task 8: Pipeline as Code and Server Deployment

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## 1. Objective

Automate application build/deployment through a Jenkinsfile and deploy the Spring Boot JAR to the Windows environment.

## 2. Deployment

```
C:\DeliveryPromise\delivery-promise-calculator-0.0.1-SNAPSHOT.jar
Application: 8765
Jenkins: 8080
```

## 3. Jenkinsfile Core

```
pipeline {
    agent any
    stages {
        stage('Checkout') {
            steps { checkout scm }
        }
        stage('Build and Test') {
            steps {
                bat '"C:\DevTools\apache-maven-3.9.16\bin\mvn.cmd" clean test package'
            }
        }
    }
}
```

## 4. Jenkins Process-Cleanup Issue

A background Spring Boot process was initially terminated by Jenkins process cleanup. The deployment was corrected by setting Jenkins process-cookie variables and launching the application as a detached process.

## 5. Deployment Command

```
set "JENKINS_SERVER_COOKIE=dontKillMe"
set "JENKINS_NODE_COOKIE=dontKillMe"
start "Delivery Promise Calculator" /B cmd /c "java -jar "C:\DeliveryPromise\delivery-promise-calculator-0.0.1-SNAPSHOT.jar"
```

## 6. Result

The pipeline successfully built the application and deployed the JAR on port 8765.

# Task 8 — Actual Project Screenshots

Screenshots captured during the implementation are included below as evidence. These are the actual project screenshots, not recreated diagrams.
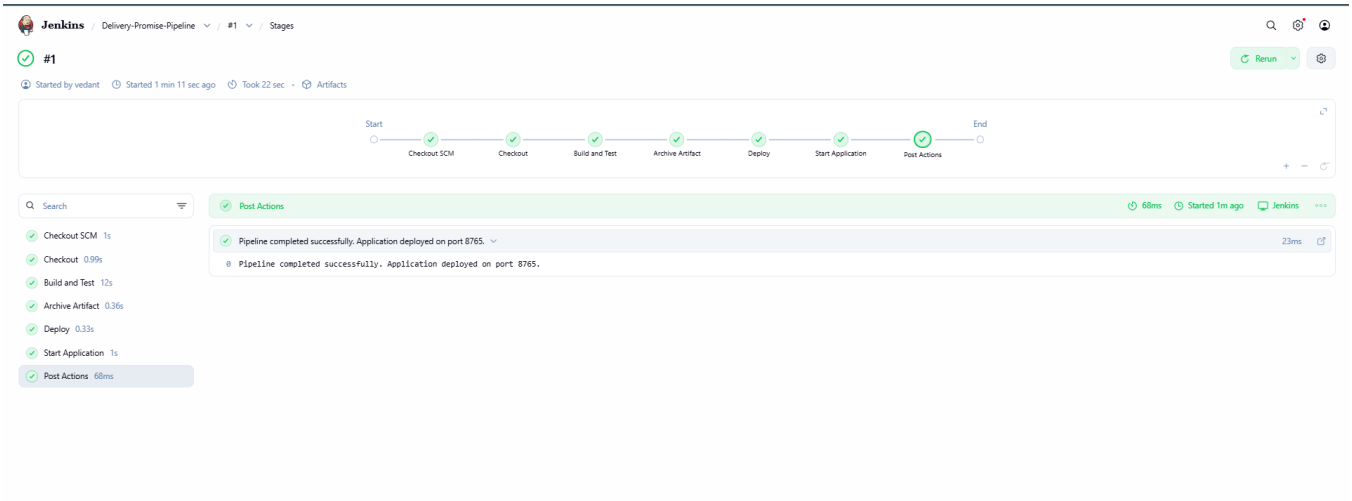
Figure 1. Jenkins pipeline showing successful checkout, build/test, artifact, deployment and application-start stages.